Can an AI Agent Replace a Sales Rep Who Already Uses AI?

A Proposal for a Multi-Firm Field Experiment in B2B Outbound Sales

Final Report (Research Proposal), Topics in Econometrics B

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1. Motivation

My bachelor thesis starts from a simple premise: what keeps a start-up small is staff. Growth needs people, and people are slow to hire and expensive to keep. AI agents promise to loosen this constraint, and the entry-level sales role is where they make that claim first. A business development associate (BDA) researches prospects, contacts them by phone, e-mail and LinkedIn, follows up and books meetings; an autonomous agent can now do the same by e-mail. Whether it does them as well decides whether a start-up still needs to hire for this role. This proposal asks:

"How much does replacing an AI-assisted BDA with an autonomous AI agent change the share of contacted prospects that book a meeting, and at what cost per prospect?"

2. Related Work and Contribution

Most field evidence on generative AI at work answers a neighboring question. In customer support, an AI assistant raised the number of issues resolved per hour by 15% on average, especially for less experienced workers (Brynjolfsson et al., 2025). In professional writing tasks, ChatGPT cut the time needed and raised quality (Noy & Zhang, 2023). Both studies treat AI as a tool in a human's hand. They measure augmentation, not replacement.

The closest study compares human and automated sales agents directly. In a randomized field experiment (n = 325) and an online experiment (n = 408), Adam et al. (2023) find that customers are more willing to show initial interest to a human sales agent, but more willing to leave their contact details with an automated one. As far as its abstract reports, the study compares automated agents with humans and with human-automated combinations, not with a human who does the whole job with AI. Both decisions it studies come before a meeting. Who makes the contact also matters in its own right: in outbound sales calls, disclosing that the caller was a chatbot cut purchase rates by 79.7% (Luo et al., 2019).

To my knowledge, no randomized study compares an autonomous agent with a human who already uses AI, on booked meetings, across more than one start-up. This proposal fills that gap, with Adam et al. (2023) as its benchmark.

3. Data: Where and How

Such a comparison needs data from more than one start-up. The experiment would run at Finto and three further B2B start-ups, each contributing one BDA for a two-week pilot and a twelve-week main phase. In return, each start-up uses the agent free of charge during the study and receives its own results. Finto, a Munich-based AI start-up where I worked over the summer, sets the baseline: of the roughly 1,000 firms I contacted there, about 100 a week, two to three in every 100 booked a meeting. The pilot rechecks this rate for the BDAs' channel mix. The pool of prospects is finite, so each start-up confirms before the start that its target profile, drawn from its CRM and a commercial prospect database, yields about 4,800 accounts: 1,200 for the BDA and 3,600 for the agent.

All outcomes come from systems the start-ups already run. The sending tool records every e-mail, bounce and reply; the dialer and the CRM record calls, LinkedIn messages and booked and held meetings with timestamps; the BDA logs working hours; the agent logs its API spend. Pre-assignment covariates (industry, firm size, country, lead source) come from the CRM and the prospect database. Each start-up exports pseudonymized account-level records under a data-sharing agreement. The BDAs consent, outreach follows the rules on unsolicited messages, and the study needs ethics approval.

4. Design and Parameter of Interest

So what should the agent be compared with? The obvious control group is a human without AI. It would be the easier comparison and would show the larger effect. It is also the wrong one. A start-up that considers replacing its BDA today compares the agent with a BDA who already drafts e-mails with ChatGPT, so only this comparison answers the replacement question. The control arm is therefore a BDA who works as usual, by e-mail, phone and LinkedIn, and may use any general-purpose AI tool but not the agent; the tools used are logged. The comparison thus bundles author and channel, which is exactly what a replacement changes.

Field experiments with sales and call-center staff usually randomize the workers (Bloom et al., 2015; Sandvik et al., 2020). Here the agent does not change how the BDA works but takes the BDA's place, so the unit of randomization is the prospect firm (the account), not the contact person; two colleagues at the same prospect then never hear from both arms (Sawada, 2026, Lecture 2). Each week, a new batch of accounts is assigned within strata of start-up × week (Bruhn & McKenzie, 2009). Because only the human arm is capacity-bound, three accounts go to the agent for every account the BDA handles (List et al., 2011). Each arm owns its accounts until a meeting is booked or the sequence ends, and the arms send from separate domains. Both rules limit interference between the arms. The agent writes only e-mails and states in each that it is an AI assistant.

The two arms hit different limits. The median sales development representative (SDR, the common US title for a BDA) logs 112 activities a day, 44 of them calls and 41 e-mails (Bertuzzi, 2025). These are touches across sequences: at an assumed eight to ten touches per account, 560 touches a week reach only 56 to 70 new accounts, so my 100 a week at Finto is an optimistic pace. The pilot therefore measures each BDA's pace, which then fixes the size of all twelve weekly batches; a researcher draws each batch from the pool and randomizes it before the BDA sees it. The agent is limited by deliverability, not time. A new mailbox needs weeks of warm-up, after which a typical mailbox sends 14 to 42 cold e-mails a day (Smartlead, n.d.), and Google's sender guidelines ask for authentication and reported spam below 0.1% (Google, n.d.). Every mailbox in either arm is therefore capped at an assumed 40 cold e-mails a day. At an assumed four e-mails per account, the agent's 300 new accounts a week need six warmed mailboxes per start-up and the BDA's 100 need two. The agent wins on volume, the BDA perhaps on quality, so both the meeting rate per account and meetings per week are measured.

The parameter of interest is the intention-to-treat effect $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$ on the accounts of the participating start-ups, where Y_i equals one if account i books a meeting within 30 days of the first contact. τ is the net policy effect of deploying an e-mail-only, disclosed agent rather than an AI-assisted, multichannel BDA, and the booked meeting is the single primary outcome. Cost is the second quantity the question names. Hours and set-up time accrue per week, not per account, so cost per account is measured for each start-up, week and arm: API spend, mailboxes and a share of supervision time for the agent, logged hours times the start-up's loaded wage for the BDA. As a benchmark, the median SDR earns a base of USD 55,000, or USD 80,000 with variable pay (Bertuzzi, 2025), and wages are 70.3% of employer costs (U.S. Bureau of Labor Statistics, 2025). That puts a BDA at roughly USD 38 to 55 per hour, or USD 15 to 22 per account at 100 accounts a week. The cost difference between the arms, τ_C, gives the net value of a meeting worth v as $v\tau - \tau_C$, reported over a range of v so that no estimate close to zero ever sits in a denominator.

5. Estimation and Power

The meeting effect is estimated by

$$Y_i = \tau A_i + \gamma_{s(i)} + (X_i - \bar{X})'\beta + A_i(X_i - \bar{X})'\delta + \varepsilon_i,$$

where A_i is assignment to the agent, $\gamma_{s(i)}$ are stratum fixed effects and X_i are the centered pre-assignment covariates, interacted with A_i as in Lin (2013). The covariates add precision, not identification. Assignment varies at the account level, and the four start-ups are not a random sample of start-ups, so heteroskedasticity-robust standard errors fit the estimand (Abadie et al., 2023). τ_C is estimated across the 48 start-up-weeks.

Three further rules are fixed in advance. Secondary outcomes (any reply, positive reply, bounce, unsubscribe) are corrected for multiple testing with the Romano-Wolf step-down procedure (Romano & Wolf, 2005; Sawada, 2026, Lecture 2, Part 2). Accounts the BDA never works stay in their arm under ITT. If a prospect bypasses the agent and reaches the BDA, assignment serves as an instrument for the arm that handled the account, which identifies the effect for compliers as long as no BDA account reaches the agent (Bloom, 1984; Imbens & Angrist, 1994; Sawada, 2026, Lecture 3). Whether a booked meeting is held is defined only for accounts that book, so comparing show-up rates would condition on a post-treatment outcome; the target is the effect on accounts that would book with either arm, bounded under monotonicity of booking (Frangakis & Rubin, 2002; Lee, 2009; Sawada, 2026, Lecture 10).

How large an effect the design can detect depends on the base rate and on the BDAs' pace. At $p_0 = 2.5\%$ and 100 accounts a week, four BDAs over twelve weeks give 4,800 human accounts, and with 14,400 agent accounts on top the design detects a rise to 3.26% (+30%) or a fall to 1.80% (−28%). At 150 accounts a week the bounds narrow to +25% and −23%; at 50 they widen to +43% and −39% (Appendix, Tables A2 and A3). A Monte Carlo check of the estimator under independent accounts agrees: its 95% intervals cover the true effect in 93 to 95% of 1,000 replications, it rejects a true null in 6.2% and detects +30% in 76 to 78% (Appendix, Table B1). A null result therefore means that changes of about 30% would probably have been found, not that the arms perform equally. The experiment runs into the very constraint it is meant to test: the number of people.

6. Threats to Validity

Size is not the only threat. The agent learns, so the treatment changes while it is being measured, which breaks the "no hidden treatments" part of SUTVA (Sawada, 2026, Lecture 2). Updates are therefore allowed only between weekly batches and are logged, so the pooled τ averages over the agent versions used. BDAs who know about the comparison may work harder, a John Henry effect that would understate τ; activity and hours are logged to check this. Because the agent only writes, a positive result shows that e-mail alone can do the BDA's multichannel job, not that an agent could also call. Finally, four participating start-ups chosen by access are not all start-ups, and each BDA is one person. The estimate holds for these start-ups and these people.

The experiment returns to the opening premise with a number. If the agent's net value per account is positive across plausible meeting values, staff stops being the constraint at the first stage of sales. If it is negative, a start-up still needs people to grow.

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Appendix

The power calculation (Tables A1 to A3) and the simulation (Table B1) were run in Python; code and results are public (Jensen, 2026). The simulation contains no estimates of the agent's effect. It only checks that the planned estimator behaves as the power calculation promises, under independent accounts.

A. Power calculation

Two-sided test of two proportions at $\alpha = 0.05$ and power 0.80, normal approximation with unpooled variance. The regression of Section 5 can only be more precise, so the calculation is conservative. The control rate $p_0 = 2.5\%$ comes from Finto. With n_0 human and n_1 agent accounts, the smallest detectable rise $p_1 > p_0$ solves

$$\frac{p_1 - p_0}{\sqrt{p_0(1-p_0)/n_0 + p_1(1-p_1)/n_1}} = z_{0.975} + z_{0.80},$$

and the largest detectable fall is found the same way for $p_1 < p_0$.

Table A1. Accounts needed per arm in a 1:1 design.

p_0	**+25%**	**+50%**	**+100%**
2.0%	13,806	3,822	1,138
2.5%	10,981	3,038	903
3.0%	9,097	2,514	746

Table A2. Detectable rise / fall by number of start-ups (one BDA each, 100 accounts per week, 12 weeks, 1:3 allocation). The main design is the row with four start-ups.

Start-ups	**Human accounts**	**Agent accounts**	$p_0 = 2.0\%$	$p_0 = 2.5\%$	$p_0 = 3.0\%$
1	1,200	3,600	+71% / −60%	+63% / −54%	+57% / −50%
2	2,400	7,200	+49% / −44%	+43% / −39%	+39% / −36%
3	3,600	10,800	+40% / −36%	+35% / −32%	+32% / −30%
4	**4,800**	**14,400**	**+34% / −31%**	**+30% / −28%**	**+27% / −26%**
6	7,200	21,600	+28% / −26%	+25% / −23%	+22% / −21%
8	9,600	28,800	+24% / −22%	+21% / −20%	+19% / −18%

Table A3. Detectable rise / fall by the BDAs' pace (four start-ups, 12 weeks, 1:3, $p_0 = 2.5\%$). The pace is measured in the pilot and sets the size of the weekly batches.

New accounts per BDA and week	**Human accounts**	**Agent accounts**	**Detectable rise / fall**
50	2,400	7,200	+43% / −39%
75	3,600	10,800	+35% / −32%
100	**4,800**	**14,400**	**+30% / −28%**
150	7,200	21,600	+25% / −23%
200	9,600	28,800	+21% / −20%

Giving the agent more accounts helps less than it seems, because the variance of the smaller human arm dominates. At $p_0 = 2.5\%$, raising the ratio from 1:3 to 1:5 (9,600 more agent accounts) lowers the detectable rise only from +30% to +28%; one more start-up (4,800 accounts) lowers it to +27%.

B. Monte Carlo check of the design

Each replication draws four start-ups with baselines that differ on the logit scale (standard deviation 0.3, population mean of p_0 calibrated to 2.5%) and accounts with three covariates standing in for those of Section 3. Outcomes are independent across accounts given these baselines; week-level shocks, non-compliance and cost are not simulated. Assignment is 1:3 within start-up × week. The agent multiplies each account's baseline probability by $(1 + r_f)$; in the heterogeneous scenarios r_f varies across start-ups with standard deviation 0.10. The true effect is computed from the simulated probabilities, so coverage is measured against the effect the estimator targets.

Two tests are checked. The **main test** is the Lin (2013) regression of Section 5 with stratum fixed effects and robust standard errors; its target is the effect on the participating start-ups. The **start-up test** averages one Lin estimate per start-up and uses a t-test with three degrees of freedom (Ibragimov & Müller, 2010); its target is the average effect across start-ups like these.

Table B1. Rejection rates and coverage of 95% intervals, 1,000 replications each.

Scenario	Main test: rejects	Main test: coverage	St
No effect, same relative effect in every start-up	6.2%	93.8%	
No average effect, varies across start-ups	8.8%	94.9%	
Effect at the detectable size (+30%), same relative effect in every start-up	77.9%	93.4%	
Effect at the detectable size, varies across start-ups	76.2%	94.8%	

Under no effect, the main test rejects in 6.2% and the start-up test in 4.5% of replications; with 1,000 replications the Monte Carlo standard error is about 0.7 points, so the main test is slightly liberal. In the second row, the main test rejects more often because the effect on the four participating start-ups is not exactly zero when effects vary across start-ups; its coverage of that effect stays at 95%. At the detectable effect, the main test comes close to the planned power (76 to 78% against 80%), while the start-up test, which answers a broader question with four units, reaches only 34 to 42%. The headline estimand is the effect on the participating start-ups because they are not a random sample (Section 5); the lower power of the start-up test is a second reason, not the first.

C. Timeline and pre-specification

Table C1. Timeline of the experiment.

Weeks	Phase	Content
before 1	Warm-up	Agent and BDA mailboxes are warmed up on separate domains, capped at 40 cold e-mails a day each; authentication (SPF, DKIM, DMARC) is set up.
1–2	Pilot	Normal BDA pace at 1:3 in each start-up to test logging and deliverability, measure the BDA's pace, which then fixes the batch size, and see how much cost per account varies across weeks. Analysis code registered before the main phase. Pilot data are not used in the main analysis.
3–14	Main experiment	12 weekly batches per start-up, drawn and randomized by a researcher at the pace measured in the pilot (100 BDA and 300 agent accounts at the planned pace). Agent updates only between batches, logged.
15–18	Outcome closure	30-day booking windows of the last batch close; meetings held are verified.
19–20	Analysis	Pre-specified estimators of Section 5 and the start-up test of Appendix B.